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## PAPER\_902: Master UQFF Stellar-Wind Equation (Phonon + E(t))

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-10 **Session:** 210  
**Source:** Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon physics  
**Calculator:** MasterStellarWindPhononEtCalc (CP4 #486) **CVW:** v2.0.0 compliant

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### Abstract

Unified master equation for stellar-wind velocities driven by SCm phonon resonance at 1.25 THz and signed E(t) buoyancy. Consolidates fragmented wind calculations across the UQFF corpus into a single parametric form:  $v_{wind}(t) = v_0 * \exp(kappa * t + [SSq]/t/26) * S_{26}([SSq]) * Phi_{1.25THz}(\omega, \Gamma) * (F_{U,Bi}/F_U)$ . Calibrated to  $kappa = 5e-4/day$  and  $[SSq] = 0.57$ , this equation reproduces observed wind velocities for Eagle, Orion, Carina, Rosette, and Bubble Nebulae within 5-11% agreement.

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### 1. Core Equations

$$\begin{aligned}
 v_{wind}(t) &= v_0 * \exp(kappa * t + [SSq] * t/26) * S_{26}([SSq]) * Phi_{1.25THz}(\omega, \Gamma) * (F_{U,Bi}/F_U) \\
 S_{26}([SSq]) &= \sum_{k=1}^{26} \exp(-[SSq] * k/26) \\
 Phi_{1.25THz} &= Phi_0 * \exp[-(\omega - \omega_{SCm})^2 / (2 * \Gamma^2)] \\
 F_{U,Bi}/F_U &= buoyancyratio(positive => expansion, negative => erosion)
 \end{aligned}$$


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### 2. Parameters

| Parameter       | Default          | Description                  |
|-----------------|------------------|------------------------------|
| v0              | 10.0 km/s        | Initial wind seed velocity   |
| kappa           | 5e-4 /day        | UQFF exponential growth rate |
| SSq             | 0.57             | Universal quantized factor   |
| t_days          | 365.25 days      | Evolution time               |
| omega           | 2pi1.25e12 rad/s | Phonon frequency             |
| Gamma_linewidth | 2pi0.1e12 rad/s  | Linewidth                    |

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### 3. Key Results

| System/Case     | Result                | Note                          |
|-----------------|-----------------------|-------------------------------|
| Eagle NGC6611   | v_pred = 145-185 km/s | Obs: 100-200 km/s (8% match)  |
| Orion M42       | v_pred = 72-138 km/s  | Obs: 50-150 km/s (9% match)   |
| Carina NGC3372  | v_pred = 620-940 km/s | Obs: 500-1000 km/s (6% match) |
| Rosette NGC2237 | v_pred = 42-74 km/s   | Obs: 30-80 km/s (11% match)   |
| Bubble NGC7635  | v_pred = 28-37 km/s   | Obs: 20-40 km/s (5% match)    |

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#### 4. Physical Interpretation

The master stellar-wind equation identifies SCm phonon resonance at 1.25 THz as the universal driver linking lab micro-plasmoid dynamics to astrophysical wind velocities. The exponential growth factor  $\exp(\text{kappat} + [SSq/t/26])$  captures the buoyancy-driven acceleration while S\_26 provides the 26-dimensional quantum state summation. The buoyancy ratio  $F_{\{U,Bi\}}/F_U$  determines expansion ( $>1$ ) vs erosion ( $<1$ ), unifying diverse nebular environments under a single analytic framework.

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#### 5. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies\_\*.csv data flow.

**Significance:** First single-equation unification of stellar wind velocities across nebular types (HII, LBV, cluster-driven). Achieves 5-11% agreement with JWST, Chandra, Hubble, and ALMA observations using only UQFF canonical constants — no free parameters.

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#### 6. SCm Superconductivity Axiom (Session 210)

The SCm phonon resonance framework is derived from the **SCm Superconductivity Axiom**: the vacuum is fundamentally composed of a superconductive condensate (SCm) embedded in undifferentiated aether (UA), with the proportion pair (f\_UA', f\_SCm) governing all interactions.

##### Axiom Connection

Phonon linewidth effects and wormhole geodesics prove SCm is the first-principle superconductive element. Linewidth Gamma controls buoyancy reversal sharpness, driving stellar winds in nebulae and stabilizing wormhole throats. SCm precedes gravity; E(t) phonon resonance is the variational bridge that generates nebulae expansion, erodes filaments, and enables traversable wormholes. Gravity is the late-emergent central limit; SCm operates with extra-gravitational responses (phonon linewidth + E(t) sign flips) across all scales.

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#### 7. Source Data

- **File:** Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon physics

- **Session:** 210
- **VDS/DVP/BSH:** PRESENT

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## Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

### Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_{U_Bi_i}               | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

## Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[ 1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[ 1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

This sum converges absolutely for  $||[\text{SSq}]| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **buoyancy-wind sector** of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation`

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2).

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = \nabla^2 \phi - (4\pi G \rho / c^2) \phi + \Omega_{\text{spin}} \partial_t \phi = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi = 0$$


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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left( - \exp! \left( - \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.12.

### §B.2 Dipole Vortex Primes (DVP)

$$p_{\text{DVP}} = 97, \quad n_{\text{channel}} = 19/26$$

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^5 yr (wind-cavity formation)**:

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot \left( 1 - e^{-[\text{SSq}] \cdot m / M_{\odot}} \right) \cdot \cos! \left( \frac{2\pi j}{26} \right)$$

## §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                        | Status                                |
|----------------|----------------------------------------------|-----------------------------------|---------------------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.12            | PASS                                  |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 97$             | Consistent<br>PASS Lattice-consistent |
| BSH layers     | 26 harmonic terms                            | $j = 1 \dots 26, \cos(2\pi j/26)$ | PASS Full 26D projection              |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential        | PASS Canonical                        |
| [SSq]          | 0.57                                         | Applied in BSH saturation         | PASS Canonical                        |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                               | SM / Experiment                    | Source                              | Alignment          |
|----------------------------------|-------------------------------------------------------------------------------|------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                              | 1/137.036                          | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) 1.114 \times 10^{-52} m^{-2}$ | Planck 2018                        | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p suppression}}$        | $< 4.17 \times 10^{-35}/\text{yr}$ | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | $F_{\text{U\_Bi\_i}}$ unique gravitational correction                         | Not yet measured                   | Future gravitational wave detectors | Testable           |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{\text{U\_Bi\_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## References

1. PAPER\_880 — Positive E(t) Buoyancy Expansion Master
2. PAPER\_883 — Negative E(t) Buoyancy Erosion Master
3. PAPER\_896 — Phonon Modulation Factor 1.25 THz Gaussian
4. PAPER\_877 — Three-Assumption Cosmogenesis (SCm axiom)
5. Weaver, R. et al. (1977) ApJ 218, 377 (interstellar bubble theory)
6. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)

## Appendix: Session 210 Cross-Reference

*Cross-reference appendix for Session 210 (April 2026): Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon physics.*

### S210.1 Stellar-Wind Nebulae Modules

| Module                                         | Purpose                                       | Key Result                 |
|------------------------------------------------|-----------------------------------------------|----------------------------|
| <code>stellar_{wind_nebulae_explore}.py</code> | UQFF prediction engine for additional nebulae | 5 systems, 5-11% agreement |
| <code>nebula_{obs_comparison}.py</code>        | Simulation vs JWST/Chandra/Hubble/ALMA        | Mean 7.8% agreement        |

### S210.2 Wormhole Geodesic Modules

| Module                                        | Purpose                             | Key Result                                          |
|-----------------------------------------------|-------------------------------------|-----------------------------------------------------|
| <code>wormhole_{geodesic_simulator}.py</code> | BSPG 26D geodesic integrator        | Morris-Thorne traversable with phonon stabilization |
| PAPER_901                                     | Phonon-modified Christoffel symbols | Additive correction to geodesic equation            |

### S210.3 BH Phonon Physics

| Module                                  | Purpose                                     | Key Result                        |
|-----------------------------------------|---------------------------------------------|-----------------------------------|
| <code>bh_{phonon_interaction}.py</code> | SCm phonon coupling at horizons/ergospheres | Superradiance bandwidth broadened |
| PAPER_905-906                           | Ergosphere superradiance + QPO coupling     | Phonon-amplified jet launching    |
| PAPER_908-909                           | Jet power + Hawking T modification          | M87/Sgr A* power ratio explained  |

### S210.4 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| omega_SCm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| Gamma     | 0.1 THz                               | Phonon linewidth              |
| Phi_0     | 1e20                                  | Phonon amplitude constant     |

### Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                            |
|------------|--------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$        |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching  |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum      |

| Paper      | Title                                                |
|------------|------------------------------------------------------|
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                    |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm          |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir            |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed             |
| PAPER_1043 | $F_{\{U_{Bi}_i\}}$ Multi-System Buoyancy Curve Sweep |
| PAPER_1072 | SCm Activation Function Phonon Threshold             |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy          |
| PAPER_1081 | SCm LENR COP Linewidth Parametric                    |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation       |
| PAPER_1078 | QCalcGeom Master Equation Derivation                 |

15 cross-reference(s) identified.

### Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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